DS 501 · INTRODUCTION TO DATA SCIENCE

Week 01

What Is Data Science?

What data science is, how a data science project runs, what kinds of data it uses, and who does the work.

THIS WEEK

- **1-1** Introduction to Data Science — 45 minutes
- **1-2** The Data Science Process and Types of Data — 45 minutes
- **1-3** The Data Scientist and the Data Team — 45 minutes
- **P1** Practice: a first look at real data — 60 minutes · Python notebook

International Balkan University · [Instructor name] · [Semester]

The DS 501 Story in Six Steps

Every week of the course covers one step on the path from a question to a responsible decision.

01

Ask a question that supports a decision · weeks 1–2

02

Get the data: files, APIs, databases · weeks 3–4

03

Understand the data: clean, explore, describe · weeks 5–6

04

Infer and model: statistics and machine learning · weeks 7, 9–10

05

Communicate: visualise, tell the story, scale up · weeks 8, 11

06

Act responsibly: ethics, privacy, fairness · week 12

Key point: your course project follows the same six steps, from a question you choose in week 1 to a presentation in week 12.

How the Course Works

Slides and reading notes

Each week is one deck. The notes under every slide are your reading text: read them after class.

Weekly practice

A Python notebook in Google Colab every week, started in class and submitted before the next session.

Weekly quiz

15 questions on the lesson objectives. The Key Terms slides use the exact quiz wording.

Papers to read

Two or three short papers each week. Bring a half-page note on each required paper.

Course project

One project all semester, on a question you choose. Milestones in weeks 3, 5, 7, 10 and 12.

AI tools

Allowed for help with code and writing, if you check the output and state how you used them.

Assessment weights and due dates are in the syllabus: labs [__%] · quizzes [__%] · project [__%] · participation [__%].

DS 501 · WEEK 1 · LESSON

1-1

Introduction to Data Science

Lesson 1 of 3

In this lesson

01 Data, information and knowledge

02 What data science is, and why definitions differ

03 Why the field grew so quickly, and where it came from

04 Data science, statistics, AI and big data

45 minutes

No prior knowledge needed

Objectives

By the end of this lesson, you will be able to:

01 **Distinguish** data, information and knowledge, and place examples on the DIKW hierarchy

Checked in: Quiz 1 · questions 1–2

02 **Explain** what data science is and why its definitions differ

Checked in: Quiz 1 · question 3

03 **Explain** why data science grew so quickly and outline its history

Checked in: Quiz 1 · question 4

04 **Distinguish** data science from statistics, big data, AI and machine learning

Checked in: Quiz 1 · question 5

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Objective 02

Case: “We Need Data Science”

THE SITUATION

At a regional bank, the board has read that competitors use data science. The CEO tells the head of IT:

> “We need data science. Hire a data scientist and buy whatever AI platform they want.”

Nobody in the meeting can say what problem the data scientist should solve.

YOUR TASK · PAIRS · 5 MIN

1. Write your own one-sentence definition of data science.
2. Is it statistics, programming, AI, or something else?
3. What should the bank decide **before** it hires anyone?

Objective 01

Data Versus Information

Data

Raw, recorded facts about objects, events or people.

No meaning on its own until it is put in context.

Example: **7, 8, 9, 6**; a sensor reading of 42.1; a customer ID.

Information

Data that has been **organised**, combined, compared or calculated.

Answers a question: who, what, where, when or how many.

Example: “the class average is 7.5, and **two** students are below the pass mark”.

⚠ **Common mistake:** a large table is not information by itself. Data becomes information only when it answers a question someone is asking.

Objective 01

From Data to Wisdom: The DIKW Hierarchy



Wisdom: Issue health alerts and subsidise cleaner heating before winter.

Knowledge: Peaks occur on cold, windless days with heavy household heating.

Information: PM2.5 exceeded the EU daily limit on [__] days this winter.

Data: Station Centar, 12 Jan, 08:00: PM2.5 = 87 µg/m^3.

- Each level adds **context** and **meaning** to the level below it.
- Data science helps an organisation move **up** the hierarchy, from facts to decisions.

Objective 02

Defining Data Science

- **Data science** is the study and practice of extracting knowledge from data in order to understand a phenomenon and support a decision.
- It is **interdisciplinary**: no single older field covers all the skills it needs.
- It covers the whole path from a **question** to a **decision**, not only the modelling step.
- It works with structured tables and with unstructured text, images and audio.

Four ingredients

- Statistics and mathematics
- Computing and data engineering
- Domain knowledge
- Communication

Key point: if any one of the four ingredients is missing, the result is either wrong, irrelevant or ignored.

Objective 02

How Practitioners Define Data Science

Data scientists interviewed in the IBM course describe their field in different ways:

A process

“Data science is a process, not an event”: using data to understand the world and to test ideas against evidence.

An investigation

The art of uncovering the insights and trends hidden behind data, then turning them into a story.

A science of data

As biology studies living things, data science studies data, which has real properties that must be understood.

A practice

If you have data and curiosity, and you explore and analyse the data to get answers, you are doing data science.

Haider: a data scientist solves problems with big **or small** data, using appropriate tools, and tells the story to stakeholders.

Objective 02

Data science starts with a question, not with a dataset or a tool.

Data and algorithms are only useful when they answer a question that someone needs answered. The same dataset can support many decisions, and the question decides which data, which methods and which result matter.

Rule of thumb: if you cannot name the decision a result will change, you are not ready to analyse the data.

Objective 03

Why Did Data Science Grow So Quickly?

Four changes happened at the same time after about 2010, and each one reinforced the others:

A flood of data

Web logs, phones, sensors, social media and digital payments produce continuous streams of data.

Cheap storage

The cost of storing a gigabyte fell sharply, so organisations keep history instead of deleting it.

Free, open-source tools

Python, R, pandas, scikit-learn and Spark are free and well documented, and run on a laptop or in Colab.

Computing power and algorithms

Cloud computing and GPUs make large models practical, and machine learning turns that power into predictions.

Objective 03

A Short History of Data Science

Year	Who	Contribution
1962	John Tukey	“The Future of Data Analysis”: analysis as a science of its own
1974	Peter Naur	Uses the term “data science” in a book on computing
2001	Cleveland; Breiman	An action plan for the field; the two cultures of modelling
2008	Patil and Hammerbacher	“Data scientist” as a job title at LinkedIn and Facebook
2012	Davenport and Patil	“The sexiest job of the 21st century” (Harvard Business Review)
2017	David Donoho	“50 Years of Data Science”: the six divisions
2022 →	Generative AI	Large language models become everyday tools

In practice

Tukey 1962 and Donoho 2017 are on this week's reading list.

Read the first row first. The core idea is 60 years old; what changed after 2010 was the scale.

Objective 04

Donoho: The Six Divisions of Greater Data Science

Donoho (2017) argues that data science is statistics enlarged to cover everything a data professional actually does:

01

Data gathering, preparation and exploration: collecting, cleaning and exploring data

02

Data representation and transformation: databases, file formats and features

03

Computing with data: programming, clusters, the cloud and workflows

04

Data modelling: statistical models and predictive machine-learning models

05

Data visualisation and presentation: from exploratory plots to reports

06

Science about data science: studying which methods work in practice

In practice

DS 501 covers divisions 1–5 directly: weeks 3–6 (1, 2), weeks 4 and 11 (3), weeks 7, 9–10 (4), week 8 (5).

Breiman's Two Cultures of Modelling

Breiman (2001) described two ways of learning from data. Modern data science uses both.

Data modelling culture

Assumes the data come from a known **stochastic model**, such as linear regression.

Goal: explain how the inputs relate to the output.

Judged by: goodness of fit and statistical tests.

Traditional home: statistics.

Algorithmic modelling culture

Treats the process that produced the data as **unknown**.

Goal: predict new cases accurately, e.g. with random forests.

Judged by: accuracy on data not used for training.

Traditional home: computer science and machine learning.

Key point: choose the culture from the question: **explain** a relationship, or **predict** a case.

Objective 04

Data Science and Related Fields

Term	What it is	Relation to data science
Statistics	The science of learning from data under uncertainty	Its methodological core
Big data	Data too large, fast or varied for traditional tools	A setting, not a method
AI	Systems that do tasks linked to human intelligence	Overlaps; supplies some methods
Machine learning	Algorithms that learn from examples, not rules	The main modelling toolkit
Deep learning	Machine learning with many-layered neural networks	Used for text, images and speech
Generative AI	Models that produce new text, code or images	An assistant for analysis and code

In practice

AI ⊃ machine learning ⊃ deep learning ⊃ most generative AI. Data science overlaps all four.

Critical point: data science is **not** a branch of AI. A dashboard of waiting times is data science with no AI at all.

Objective 02

Where Data Science Is Used

Healthcare

Predicting readmissions, reading medical images, planning hospital capacity.

Banking and finance

Detecting card fraud in real time, credit scoring, measuring market risk.

Retail and e-commerce

Recommendations, demand forecasts, pricing and customer segments.

Transport and logistics

Route planning, ride pricing, traffic management, predictive maintenance.

Government

Tax compliance, social programmes, official statistics, air-quality alerts.

Education

Spotting students at risk of dropping out and planning enrolment.

Your project domain can be any of these, or the sector you work in now.

Objective 01

Exercise: Data, Information or Knowledge?

Classify each statement. Work in pairs for 3 minutes.

Statement	Data	Info.	Know.
Sensor 14 recorded 87 µg/m³ at 08:00 on 12 January.			
Average PM2.5 in Centar in January was twice the December average.			
Pollution peaks on cold, windless days because heating smoke cannot disperse.			
Customer 50213 bought two items on 3 March.			
Month-to-month customers leave about three times as often as two-year customers.			

Why? Ask of each row: is it one fact, a summary, or an explanation?

Objective 04

Check Your Understanding

A hospital builds a dashboard that shows the average waiting time in each department, updated every hour from its patient records. Which statement is correct?

A It is not data science, because it uses no artificial intelligence.

B It is data science: it turns data into information that supports decisions.

C It is big data, because the dashboard is updated every hour.

D It is machine learning, because the averages change over time.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Data

Raw, recorded facts about objects, events or people.

Knowledge

An explanation of a pattern or cause that can guide action.

Data science

Extracting knowledge from data to understand and to decide.

Machine learning

Algorithms that learn from examples instead of rules.

Information

Data organised in context so that it answers a question.

DIKW hierarchy

Data, information, knowledge and wisdom as four levels.

Big data

Data too large, fast or varied for traditional tools.

Two cultures

Breiman: modelling to explain versus algorithms to predict.

Summary

In this lesson, you learned how to:

- **O1** Distinguish data, information and knowledge, and place examples on the DIKW hierarchy
- **O2** Explain what data science is and why its definitions differ
- **O3** Explain why data science grew so quickly and outline its history
- **O4** Distinguish data science from statistics, big data, AI and machine learning

Key takeaway: data science turns data into knowledge that supports a decision, and it starts with the question.

Next lesson

1-2 The Data Science Process and Types of Data

DS 501 · WEEK 1 · LESSON

1-2

The Data Science Process and Types of Data

Lesson 2 of 3

In this lesson

01 The four steps of the data science process

02 Turning a vague request into an actionable question

03 A real case: complaints at a transit authority

04 Types of data and the file formats that hold them

45 minutes

Builds on lesson 1-1: data and information

Objectives

By the end of this lesson, you will be able to:

01 **Describe** the four steps of the data science process and how they repeat

Checked in: Quiz 1 · questions 6–7

02 **Apply** the first step: rewrite a vague request as an actionable question

Checked in: Quiz 1 · question 8

03 **Classify** data as structured, semi-structured or unstructured

Checked in: Quiz 1 · question 9

04 **Compare** common file formats and choose one for a given use

Checked in: Quiz 1 · question 10 and Practice 1

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

The Data Science Process

Every data science project, small or large, moves through the same four steps.

01

Clarify the question. Which decision will the answer support, and who makes it?

02

Identify the data. Which data can answer it, where does it come from, and may we use it?

03

Analyse and model. Explore the data and look for patterns, groups and outliers.

04

Communicate. Tell the story of the result so that someone acts on it.

Key point: the process is a loop. An answer usually raises a new question, and the analysis often sends you back for more data.

Critical point: step 1 is the step most often skipped, and it decides every step after it.

Step 1: Clarify the Question

- Organisations often bring **data first** and a question later. A good data scientist starts by asking clarifying questions.
- A clear question decides which data to collect, which methods fit, and how the result will be judged.
- Curiosity matters here: keep asking “why?” until the **business need** is clear.

Questions to ask the sponsor

- Who makes the decision, and what is it?
- What would you do differently if you knew the answer?
- Which number should change, and by how much?
- What are the deadline, budget and legal limits?

⚠ **Common mistake:** starting with whatever data is in the database. The data available should not decide the question.

Vague Questions Versus Actionable Questions

An actionable question names a **decision**, a **target** and a **time frame**.

Vague question	Actionable question	Decision
What does our sales data say?	Which product categories will miss their target next quarter, and by how much?	Adjust promotions and stock
Can we use AI for students?	Which first-year students are at risk of failing two or more courses by week 6?	Offer tutoring early
Is air quality bad in Skopje?	On which days next week will PM2.5 exceed the EU daily limit?	Issue health alerts
Why are customers unhappy?	Which three complaint types grew most since the price change in March?	Fix the main causes first

In practice

Your project proposal in week 7 must state its question in this form.

Read the middle column. Each question can be answered with data, and each answer changes what someone does.

Step 2: Identify the Data

- Once the question is clear, ask which data can answer it and **where** that data will come from.
- Data may be **internal** (transactions, customer records, logs) or **external** (open government data, weather, APIs).
- Check **quality** and **permission**: is the data complete and recent enough, and may we legally use it?

Five checks for every source

Relevance: does it measure what the question needs?

Type: table, JSON, text, images or audio?

Quality: complete, accurate, consistent, recent?

Access: who owns it, and how do we get it?

Permission: is its use legal and ethical (GDPR)?

In practice

For North Macedonia, the State Statistical Office and the national open data portal publish free datasets you can use in your project.

Steps 3 and 4: Analyse, Then Communicate

3 · Analyse and model

Explore first: distributions, summaries, missing values and simple charts.

Look for **patterns** (trends), **archetypes** (typical groups) and **outliers** (unusual records).

Try several techniques; expect to go back for more or better data.

4 · Communicate the results

Start with the answer to the business question, not with the method.

Use clear charts, state the recommended action and its **uncertainty**.

Adapt the story to the audience: executives, experts or engineers.

Key point: findings that are not communicated have no impact. Communication is part of the method, not an extra.

Objective 02

Case: Why Do Transit Riders Complain?

THE SITUATION

The Toronto Transit Commission (TTC) is one of the largest transit authorities in North America. It gave Murtaza Haider about **500,000 complaints**: structured records plus e-mails and faxes.

> "Why do some days have far more complaints than others?"

After a month of analysis, the internal data showed no clear pattern.

YOUR TASK · PAIRS · 6 MIN

1. Suggest two possible causes of the peak days.
2. Which data would test each cause? Is it in the TTC's records?
3. If the cause cannot be controlled, is the answer still useful? Why?

Lessons From the TTC Case

- **Curiosity** and everyday experience produced the key hypothesis, not a more complex algorithm.
- The answer needed **external data** (weather), so step 2 had to be revisited.
- Combining **structured** records with **unstructured** text gave a fuller picture.
- An uncontrollable cause is still **useful knowledge**: it changes staffing and communication.

> “The good news: I know why. The bad news: there is nothing you can do about it.”
>
> Murtaza Haider, on the TTC complaints analysis (IBM course)

In practice

Weather, public holidays and official statistics are free external sources that often explain patterns in internal data.

Objective 02

Start with the question, not with the data.

The question decides which data to collect, which method fits and how success is measured. Data collected without a question rarely answers the one that matters.

Rule of thumb: if you cannot name the decision, you are not ready to collect data.

Three Types of Data Structure

Structured

A **fixed schema**: rows and columns with defined types.

Stored in relational databases and spreadsheets.

Examples: bank transactions, student grades, sensor readings.

Semi-structured

Organised by **tags and keys**, but no fixed schema.

Records can be nested and can differ from each other.

Examples: JSON from an API, XML invoices, e-mail headers.

Unstructured

No predefined model or organisation.

Stored as files or in NoSQL and object stores.

Examples: e-mails, reviews, images, audio, video, PDFs.

⚠ **Common mistake:** quoting “80% of all data is unstructured” as a measured fact. It is an industry estimate; use it only as an order of magnitude.

Objective 03

Working With Unstructured Data

- Most analysis methods need structure, so much of the work is **extracting structure** from unstructured data.
- After extraction, the result is usually stored as a **table** and analysed like any structured data.
- Machine learning, and especially deep learning and large language models, has made extraction far more accurate.

From unstructured to a table

- Text → topics, entities, sentiment scores
- Images → labels, detected objects
- Audio → transcripts, then text analysis
- PDFs → extracted tables and fields

⚠ **Common mistake:** assuming automatic extraction is error-free. Check a sample of automatic labels against labels assigned by people.

Common File Formats

Data professionals choose a format by its structure, its size and who will read it.

Format	Structure	Strength	Limitation
CSV / TSV	Delimited text, one record per line	Readable by almost every tool	No data types; commas in text
XLSX	Workbook of sheets and cells	Familiar to business users	Formatting mixed with data
JSON	Nested keys, objects and arrays	Standard for web APIs	Must be flattened for tables
XML	Self-describing nested tags	Strict, validated exchange	Verbose; slow to parse
Parquet	Compressed columns, binary	Fast and small; keeps types	Not human-readable

Critical point: CSV keeps no data types; Parquet does. Dates and codes such as “007” can change silently in CSV.

Objective 04

The Same Record in Two Formats

```
# The same student record, twice
1  # students.csv
2  id,name,programme,courses
3  1042,Ana Petrova,MSc Data Science,"DS 501;CS 510"
4  # students.json
5  {"id": 1042, "name": "Ana Petrova",
6   "courses": ["DS 501", "CS 510"]}
```

2 The CSV header names the columns; every row must have the same columns.

← **3 · Critical:** a list inside one CSV cell must be split again before analysis. Either repeat rows or use a second table.

5–6 JSON keeps the courses as a real nested list.

</> **In practice**

pandas reads both: **pd.read_csv** for CSV and **pd.json_normalize** to flatten nested JSON into columns.

Choosing a File Format

- **Who will read it?** A business user needs Excel or CSV; a web developer expects JSON.
- **How big is it?** Millions of rows favour a compressed, columnar format such as Parquet.
- **Is it nested, and must types be kept?** Nested records fit JSON; Parquet and databases keep types.
- Keep the **raw file** unchanged, and save cleaned versions separately.

Write it down

Document the format, the delimiter and the **encoding**, for example UTF-8.

UTF-8 handles Macedonian Cyrillic, Albanian and Turkish characters without errors.

⚠ **Common mistake:** opening a CSV in Excel, saving it, and sharing the result. Excel may change dates, drop leading zeros and switch the encoding.

Exercise: Classify the Data

Work alone for two minutes. Mark each item as structured, semi-structured or unstructured.

Data item	Structured	Semi-structured	Unstructured
Table of library loans in PostgreSQL			
Currency-exchange API response in JSON			
Video from a traffic camera			
Excel sheet of monthly electricity readings			
Customer reviews on a shop's website			
XML invoice exchanged between companies			

Note: one row mixes two types. Find it and say which part is which.

Check Your Understanding

A weather service returns its forecast as **JSON**: one object per city, with a nested list of hourly temperatures and keys that vary between cities. How is this data best classified?

A Structured, because it comes from a database

B Semi-structured, because it is organised by keys but has no fixed schema

C Unstructured, because it is text

D Structured, as soon as it is downloaded to a file

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Data science process

Four repeating steps: clarify the question, identify the data, analyse, communicate.

Schema

A description of the structure of data: its fields and their types.

Semi-structured data

Data organised by tags or keys but with no fixed schema, e.g. JSON.

CSV

Plain-text table: one record per line, values separated by commas.

Actionable question

A question that names a decision, a target and a time frame.

Structured data

Data with a fixed schema that fits in rows and columns.

Unstructured data

Data with no predefined model, e.g. text, images, audio.

Parquet

A compressed, columnar file format that keeps data types.

Summary

In this lesson, you learned how to:

- **O1** Describe the four steps of the data science process and how they repeat
- **O2** Apply the first step: rewrite a vague request as an actionable question
- **O3** Classify data as structured, semi-structured or unstructured
- **O4** Compare common file formats and choose one for a given use

Key takeaway: start with a decision-oriented question, then choose data and formats that can answer it.

Next lesson

1-3 The Data Scientist and the Data Team

DS 501 · WEEK 1 · LESSON

1-3

The Data Scientist and the Data Team

Lesson 3 of 3

In this lesson

01 The skills and traits of an effective data scientist

02 Many paths into data science

03 The roles on a modern data team

04 What employers look for, and your own plan

45 minutes

Builds on lessons 1-1 and 1-2

Objectives

By the end of this lesson, you will be able to:

01 **Describe** the technical skills and personal traits of an effective data scientist

Checked in: Quiz 1 · questions 11–12

02 **Differentiate** the roles on a modern data team

Checked in: Quiz 1 · question 13

03 **Explain** what employers look for when recruiting for data science

Checked in: Quiz 1 · question 14

04 **Plan** your own skill development for a data science career

Checked in: Quiz 1 · question 15 and the job-posting task

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Objective 04

Case: “Can I Become a Data Scientist?”

THE SITUATION

Ana has a degree in economics and works as a junior analyst at a bank in Skopje. She uses Excel every day but has never written code.

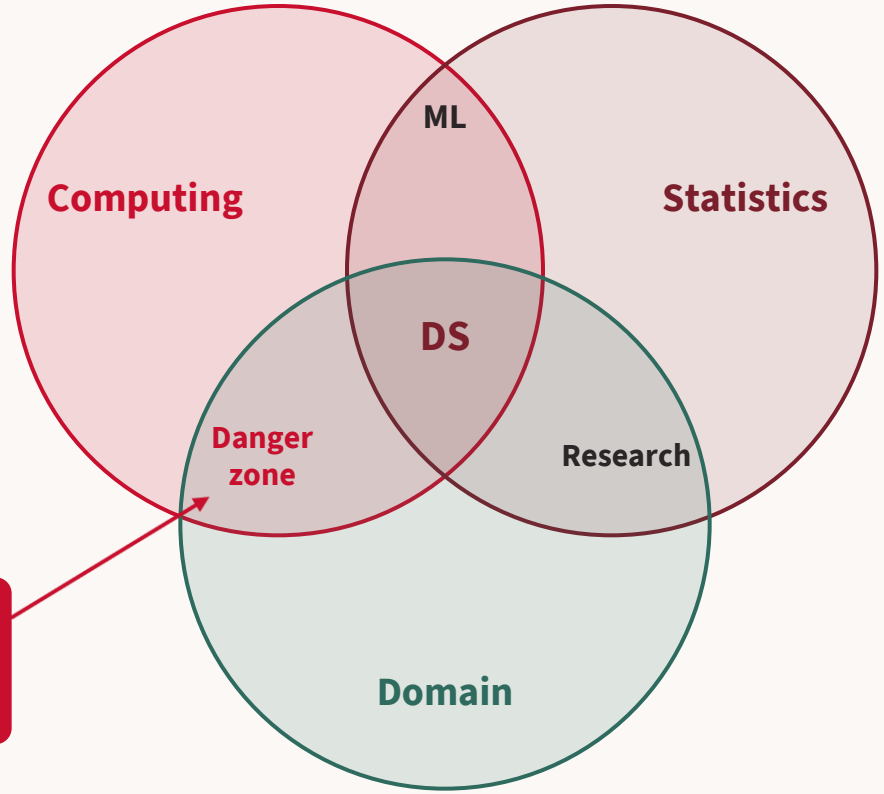
“My degree is not in computer science. Can I become a data scientist, and what should I learn first?”

YOUR TASK · PAIRS · 5 MIN

1. Which useful skills does Ana already have?
2. Which two skills should she learn first, and why?
3. What could make her stand out from computer science graduates?

Objective 01

The Data Science Skill Set



Critical point: convincing code with no statistics can be confidently wrong.

- Computing + statistics = machine learning
- Statistics + domain = traditional research
- Computing + domain, no statistics = the **danger zone**
- All three = **data science**

Technical Skills

- Effective data scientists combine **programming** and **data access** with **statistics**, machine learning and visualisation.
- Norman White's advice in the IBM course: start with programming, algebra, basic statistics and relational databases, and **walk before you run**.
- You do not need to master everything at once; depth comes from working on real projects.

Where DS 501 teaches each skill

- Python and pandas · weeks 3–5
- SQL and databases · week 4
- Probability and statistics · weeks 6–7
- Visualisation · week 8
- Machine learning · weeks 9–10
- Big data with Spark · week 11

⚠ **Common mistake:** collecting tools instead of foundations. Depth in Python, SQL and statistics is worth more than a long list of libraries.

Objective 01

Traits: Curious, Judgmental, Argumentative

Murtaza Haider's advice to new data scientists in the IBM course:

Curious

Keeps asking **why** and wants to know how things work.

Without curiosity you do not know which questions to ask the data.

Judgmental

Starts with a **hypothesis**, a preconceived idea to test.

A hypothesis tells you where to begin looking.

Argumentative

Can take and defend a **position**.

The data then shows whether the position should change.

⚠ **Common mistake:** here “judgmental” means forming a hypothesis to test, not judging people. A good data scientist changes position when the evidence says so.

Objective 01

Storytelling and Communication

- When the analysis is finished, the data scientist must **tell a story** with it; otherwise the findings stay buried.
- A good story **synthesises** a large table into one message, **surprises** the audience, and **empowers** them to act.
- Start with the answer to the business question, not with the method or the code.
- Many hiring managers rank communication as highly as technical skill.

> A great presentation is like turning a sharp corner on a mountain road and suddenly seeing the whole valley.
>
> Murtaza Haider, IBM course “What is Data Science?” (paraphrased)

In practice

DS 501 assesses communication directly: your project ends with a report and a presentation, and week 8 is devoted to data storytelling.

Objective 04

Your advantage is not the algorithm. It is the domain you understand.

Murtaza Haider's advice in the IBM course: choose a field where you know more than most people, then learn the tools that field uses. Algorithms are available to everyone; deep knowledge of a domain is not.

Rule of thumb: name the domain you know best before you list the tools you want to learn.

Objective 04

Many Paths into Data Science

- The title “data scientist” became common only around **2008–2012**, so most experienced practitioners came from other fields.
- Practitioners in the IBM course came from mathematics, economics, engineering and retail analytics.
- Your previous degree is an **asset**: it gives you a domain and a way of thinking.

Paths described in the IBM course

- Mathematics, then a statistics degree
- Business and economics, then business analytics
- Mechanical engineering, via consulting
- Civil engineering, via transport models
- Retail analysis of point-of-sale data

Key point: there is no single right background; what matters is combining the one you have with the skills you add.

A Day in the Life of a Data Scientist

Three examples from practitioners in the IBM course:

A recommendation engine

A recommender for a large organisation. The solution she was proudest of was simple, easy to explain, and as effective as a complex one.

Predicting algae blooms

Traditional methods could not predict toxic algae blooms, so a team used neural networks to warn water-treatment companies in advance.

Teaching with data

Norman White's “Dealing with Data” class at NYU Stern: Unix, Python, databases, pandas and big data, each student on a cloud machine.

In practice

Practitioners in the IBM course describe getting and preparing data as a larger part of their work than modelling.

Objective 02

Roles on a Modern Data Team

Role	Typical tasks	Typical tools
Data analyst	Reports, dashboards, answering business questions about what happened	SQL, Excel, Power BI, Tableau
Data engineer	Pipelines, warehouses and data-quality checks	SQL, Python, Spark, Airflow
Data scientist	Experiments, statistical analysis, predictive models	Python or R, scikit-learn
ML engineer	Deploying, scaling and monitoring models	Python, Docker, MLOps tools
Analytics translator	Framing questions, prioritising projects, adoption	Domain knowledge, communication

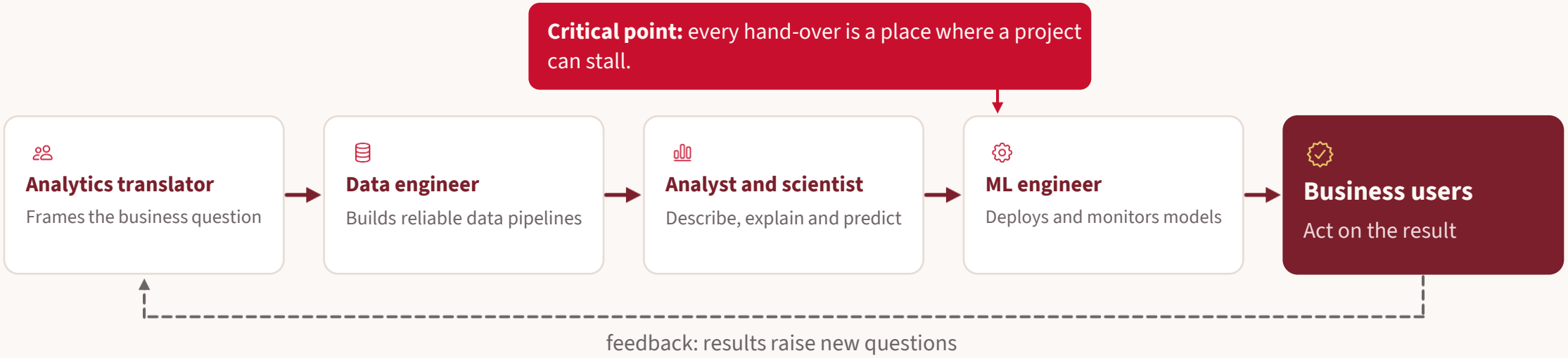
In practice

Job titles vary between companies; in a small organisation one person may cover three of these roles.

← **Read the engineer row first.** Without reliable pipelines, every other role waits for data.

Objective 02

How the Roles Work Together



- Each role owns part of the data science process, from the question to the result in use.
- Results raise new questions, so the work forms a **cycle**, not a straight line.

Recruiting: The Unicorn Problem

- Many companies look for one person with domain knowledge, data skills and great storytelling: a **unicorn** who rarely exists.
- Haider's hiring priorities put **curiosity** and **storytelling** before technical skills, because tools can be taught.
- Other practitioners look for someone relatable who builds relationships across departments.

Haider's priorities, in order

1 Curiosity about everything, not only data

2 A sense of humour and a light touch

3 The ability to see and tell a story

4 Technical skills, last: they can be taught

Key point: a portfolio of real projects shows curiosity and storytelling in a way a CV cannot.

Objective 04

Task: Read a Real Job Posting

THE TASK

Find one data science job posting that interests you on LinkedIn, Indeed, Glassdoor or a local job board.

Treat it as a dataset about your future: what does the employer really need?

> “Job postings are the most honest checklist of the skills a role needs.”

RECORD · INDIVIDUAL · 10 MIN

1. Company, job title, location, and salary range if given.
2. One responsibility linked to this week's lessons.
3. Two required technical skills, and whether you have them today.
4. Which DS 501 weeks will close your gaps?

Scenario: Who Owns This Task?

Mark the role most likely to own each task. Work in pairs for 4 minutes.

Task	Analyst	Engineer	Scientist	ML eng.
Build a nightly pipeline into the warehouse				
Create a monthly sales dashboard for managers				
Design and analyse an A/B test of two web layouts				
Put a churn model behind an API and monitor it				
Forecast demand for each product				
Fix duplicate customer records from two systems				

Why? Some rows have two defensible answers. Decide who should **own** the task, not who could help.

Objective 02

Check Your Understanding

A churn model works well in a data scientist's notebook, but it does not run in production and nobody monitors its accuracy. Which role's work is missing?

A Data analyst

B Data engineer

C Machine learning engineer

D Analytics translator

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Data analyst
Describes what happened using queries, reports and dashboards.

Data scientist
Uses statistics and machine learning to explain and predict.

Analytics translator
Links business needs and the data team; frames questions.

Danger zone
Computing plus domain knowledge without statistics.

Data engineer
Builds and maintains the pipelines and storage that make data reliable.

Machine learning engineer
Deploys, scales and monitors models in production.

Domain knowledge
Understanding of the field the data comes from.

Unicorn
An ideal candidate with every data skill at once; rarely exists.

Summary

In this lesson, you learned how to:

- **O1** Describe the technical skills and personal traits of an effective data scientist
- **O2** Differentiate the roles on a modern data team
- **O3** Explain what employers look for when recruiting for data science
- **O4** Plan your own skill development for a data science career

Key takeaway: data science is a team effort: combine technical skills with curiosity, domain knowledge and storytelling.

Next

Practice 1 · A first look at real data

Practice 1: A First Look at Real Data

You open your first notebook, load a real dataset and describe it before asking it any question.

Goal

Describe a dataset: its size, variable types, missing values and summary statistics, and compare groups.

Dataset

Palmer Penguins: **344** penguins of three species from three Antarctic islands, **7** columns.

Notebook

W01_Lab_First_Look_at_Data.ipynb in Google Colab. No installation is needed.

Time

60 minutes in class; finish the exercises at home before the week 2 session.

Parts A and B

Load and describe the data, find missing values, compute group statistics and draw two plots.

Part C

Save the same data as CSV, JSON and Parquet, and compare size and data types.

Exercises are marked ✏️. Write every answer in your own words in the notebook's text cells.

Loading and Inspecting the Data

```python
# Python · Google Colab
import pandas as pd
BASE = "https://raw.githubusercontent.com/mwaskom/"
df = pd.read_csv(BASE + "seaborn-data/master/penguins.csv")
print(df.shape)          # (344, 7)
df.info()
df.isna().sum()
```

1 Import pandas, the main Python library for tables.

2–3 Read the CSV file from its public URL into a DataFrame.

4–5 Check the size and the type of every column.

← **6 · Critical:** count the **missing values** before you compute any average.

⚠ **Common mistake:** calling **df.head()** and assuming the first five rows represent the whole table.

Comparing Groups

```python
# Python · Google Colab
df.describe().round(2)
df.groupby(["species", "sex"])["body_mass_g"].median()
df["bill_length_mm"].corr(df["bill_depth_mm"])   # -0.24
for sp, g in df.groupby("species"):
    print(sp, round(g["bill_length_mm"].corr(g["bill_depth_mm"]), 2))
```

1. Summary statistics for every numeric column.
2. Median body mass for each species and sex.
3. Correlation of bill length and depth, all penguins.

← **4–5 · Critical:** compare these signs with line 3, then explain the difference in your notebook.

In practice

Before you report a relationship, check whether a grouping variable such as species, region or customer type changes it.

Part C

One Table, Three File Formats

```python
# Python · Google Colab
df.to_csv("penguins.csv", index=False)
df.to_json("penguins.json", orient="records", indent=1)
df.to_parquet("penguins.parquet", index=False)
import os
for f in ["penguins.csv", "penguins.json", "penguins.parquet"]:
    print(f, round(os.path.getsize(f) / 1024, 1), "kB")
```

1. CSV: plain text, one row per line.
2. JSON: one object per penguin, with key–value pairs.
3. Parquet: a compressed, columnar binary file.

← **5–6 · Measure:** record each file's size, then read each file back and check the column types.

In practice

Web APIs return nested JSON; **pd.json_normalize** flattens it into a table. You will use it again in week 3.

What to Submit

Submit the completed notebook (.ipynb) on the course page. It must run from top to bottom without errors.

Part	What is marked	Points
Run	The notebook runs from top to bottom (Restart and run all)	2
A	Column types, units and missing values described in your own words	2
B	Group statistics and both plots, with the group comparison explained	3
C	File sizes recorded; a format chosen for each reader, with a reason	2
Q	One question the data could answer, phrased as a decision	1
	Total	10

⚠ **Common mistake:** pasting code output without a sentence that interprets it. Every number needs a sentence.

← **Due:** [date], before the week 2 session.

WEEK 1 · PAPERS TO READ

Papers to Read

Two classic papers on what data science is and how it differs from traditional statistics, and a method for reading any research paper efficiently.

Week 1 only

How to Read a Research Paper

Keshav (2007) suggests reading a paper in up to three passes, each one deeper than the last.

01

First pass · 5–10 min: read the title, abstract, introduction, headings and conclusions. Decide whether to read on.

02

Second pass · up to 1 hour: read carefully, study the figures and tables, skip proofs, and mark references you do not know.

03

Third pass · several hours: rebuild the argument yourself and challenge every assumption. Needed only for papers central to your work.

Key point: after the first pass, you should be able to answer the **five Cs**: category, context, correctness, contributions and clarity.

Required

Required Reading for Week 2

Read the listed sections before the week 2 class. One question in Quiz 2 may refer to these papers.

50 Years of Data Science
Donoho · Journal of Computational and Graphical Statistics 26(4), 2017

Six divisions of “greater data science”. Read sections 1–4 and 8.

Statistical Modeling: The Two Cultures
Breiman · Statistical Science 16(3), 2001

Explaining data versus predicting it. Read sections 1–5 and 11.

Bring a half-page note per required paper: the problem, the key idea, and its link to this week.

Optional

Optional Reading

The Future of Data Analysis
Tukey · Annals of Mathematical Statistics 33(1), 1962

The founding call for a science of data analysis. Read section 1.

Data Science: An Action Plan for Expanding the Technical Areas of the Field of Statistics
Cleveland · International Statistical Review 69(1), 2001

An early curriculum for data science. Six pages.

Data Scientist: The Sexiest Job of the 21st Century
Davenport & Patil · Harvard Business Review, October 2012

How the job title became popular. A short magazine article.

How to Read a Paper
Keshav · ACM SIGCOMM Computer Communication Review 37(3), 2007

The three-pass method in two pages.

Optional papers are not examined. Choose one that relates to your project.

Required

Greater Data Science Has Six Divisions

Donoho, D. (2017). 50 years of data science. Journal of Computational and Graphical Statistics, 26(4), 745–766. https://doi.org/10.1080/10618600.2017.1384734

THE PROBLEM

Around 2015 many universities launched data science programmes, often with little role for statisticians. Donoho asks whether data science really differs from statistics, and what a lasting science of data would include.

KEY IDEAS

01 Big data, new skills and job demand alone do not separate data science from statistics

02 Tukey (1962), Chambers and Cleveland long asked statistics to grow beyond mathematical theory

03 Greater data science covers every step of learning from data, grouped into six divisions

Key term: Greater data science — Donoho's name for the full field of learning from data, from gathering and preparing data to studying methods

Required

Modelling Is Only One of Six Divisions

WHAT TO TAKE AWAY

1. University teaching focuses on data modelling, the division that is easiest to formalise and teach
2. Practitioners spend more time exploring and preparing data than modelling, yet few courses teach it
3. Shared scripts and workflows, e.g. in R, turned data analysis into something that can be studied

THIS WEEK

Lesson 1-1 presents his six divisions. In Practice 1 you work in the first three: you explore the penguins, count missing values, and save the table as CSV, JSON and Parquet.

LIMITATION

Began as a talk and argues a position. Its claims about how practitioners spend their time rest on common estimates, not new measurements.

For your half-page note: Name the six divisions. Which one does Donoho say university teaching covers best, and which does he say matters most in practice?

Required

Two Cultures: Data Models and Algorithmic Models

Breiman, L. (2001). Statistical modeling: The two cultures (with comments and a rejoinder by the author). Statistical Science, 16(3), 199–231. https://doi.org/10.1214/ss/1009213726

THE PROBLEM

Most statisticians start by assuming a stochastic model for how the data were generated. Breiman, back at a university after 13 years of consulting, asks whether this habit leads to wrong conclusions and missed problems.

KEY IDEAS

01 Data modelling culture: assume a stochastic model, e.g. linear regression, and check its fit

02 Algorithmic culture: treat nature's process as unknown; judge models by accuracy on new data

03 Conclusions from a data model are about the model's mechanism, not necessarily nature's

Key term: Predictive accuracy — How well a model predicts cases not used to build it, estimated with a test set or cross-validation

Required

Judge Models by How Well They Predict New Data

WHAT TO TAKE AWAY

1. Goodness-of-fit tests and residual plots rarely detect a poor model when there are many variables
2. Rashomon effect: different models can fit almost equally well yet tell different stories
3. On hepatitis data, random forests predicted better than logistic regression and still ranked variables

THIS WEEK

Lesson 1-1's two cultures slide comes from this paper. Breiman's advice to live with the data before modelling is what Practice 1 does: describe the penguins before any model.

LIMITATION

One-sided by design. In the published discussion, Cox calls its view of statistics partly a caricature; Efron says it overstates prediction.

For your half-page note: Breiman says a model's conclusions are about the model, not nature. What does he mean, and why does he trust test-set accuracy?

Optional

Data Analysis Is a Science, Not Mathematics

Tukey, J. W. (1962). The future of data analysis. The Annals of Mathematical Statistics, 33(1), 1–67. https://doi.org/10.1214/aoms/1177704711

THE PROBLEM

In 1962, mathematical statistics focused on proofs and optimal procedures. Tukey, a leading statistician, asks whether this serves people who actually analyse data, and what data analysis should become.

KEY IDEAS

01 Data analysis is wider than inference: it also covers planning data collection and interpretation

02 It is a science, judged by experience with real data, not by mathematical proof

03 Judgment is essential: theory should guide data analysis, not command it

Key term: Data analysis — Methods to analyse data and interpret results, ways to plan data gathering, and the statistics that helps

Optional

Better an Approximate Answer to the Right Question

WHAT TO TAKE AWAY

1. Warns that chasing optimal methods under narrow assumptions could leave statistics a dried-up field
2. Asks statisticians to look at how data is actually analysed and to face messy, realistic data
3. Says teachers of data analysis should admit what the field does not know, as other sciences do

THIS WEEK

Lesson 1-1's timeline starts with this paper. His maxim, an approximate answer to the right question, matches lesson 1-2's first step: clarify the question.

LIMITATION

Long (67 pages) and written for mathematical statisticians. After the opening part, it is mostly technical methods, hard for beginners.

To think about: Tukey says data analysis must accept being wrong moderately often. Why might a science accept this, while mathematics cannot?

Optional

A Plan to Expand Statistics Into Data Science

Cleveland, W. S. (2001). Data science: An action plan for expanding the technical areas of the field of statistics. International Statistical Review, 69(1), 21–26. https://doi.org/10.1111/j.1751-5823.2001.tb00477.x

THE PROBLEM

University statistics departments worked mainly on models, methods and theory. Cleveland, at Bell Labs, asks how a department should divide research and teaching so that its work helps the practising data analyst.

KEY IDEAS

01 Judge each technical area by how much it helps the analyst learn from data, directly or indirectly

02 Six areas: multidisciplinary work, models and methods, computing, pedagogy, tool evaluation, theory

03 Gives each area a share of resources and courses, e.g. 25% for data analysis with other fields

Key term: Computing with data — Databases, software, hardware and algorithms that make models and methods practical for analysts to use

Optional

Computer Science Joins Mathematics as a Foundation

WHAT TO TAKE AWAY

1. Analysing real data in other fields is the biggest source of new tools, so students must analyse data
2. Statisticians need computer scientists' knowledge of computing; data mining needs data analysis skill
3. Data science is practised by statisticians and subject-matter analysts alike, not statisticians alone

THIS WEEK

Lesson 1-1's timeline lists this plan. Its view that subject experts also do data science supports lesson 1-3: your domain knowledge is an advantage on a data team.

LIMITATION

A six-page plan, not a study: the percentages are proposals, and the paper does not test whether the plan would work.

To think about: Cleveland gives 15% of resources to pedagogy and 5% to tool evaluation. Why should a data science department study these?

Optional

The Data Scientist as a New, Scarce Business Role

Davenport, T. H., & Patil, D. J. (2012). Data scientist: The sexiest job of the 21st century. Harvard Business Review, 90(10), 70–76. https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century

THE PROBLEM

Companies were collecting large amounts of new data but had few people who could make discoveries from it. The authors describe this new role and advise managers on how to find, attract and keep such people.

KEY IDEAS

01 A data scientist makes discoveries in big data and helps turn them into products or decisions

02 The work mixes coding, analysis, communication with the business and advice to leaders

03 Curiosity matters most; data scientists come from varied fields, not only statistics or computing

Key term: Data scientist — A professional with the training and curiosity to make discoveries in big data and turn them into value

Optional

Firms Need Data Scientists but Few Exist Yet

WHAT TO TAKE AWAY

1. At LinkedIn, Goldman's People You May Know feature showed how data analysis can become a product
2. People with this mix of skills were scarce, so firms had to recruit beyond the usual channels
3. To keep data scientists, firms must give them real problems and freedom to experiment

THIS WEEK

Supports lesson 1-3. Compare its picture of one person who does everything with the unicorn problem and the separate roles on a modern data team.

LIMITATION

A magazine article, not research: it relies on examples, and its title is promotional. Patil was himself a leading data scientist.

To think about: Which claims in this article could you check with data, for example with job postings in North Macedonia or the region?

Discussion

Reading Questions for Discussion

- Donoho names six divisions of greater data science. Which division did your previous degree cover best, and which least?
- Breiman describes two cultures. For a bank deciding on loans, would you need the data modelling culture, the algorithmic culture, or both? Why?
- Donoho and Breiman are both statisticians. Do they agree that data science is more than statistics? Where do they differ?
- Tukey wrote in 1962. Which of his ideas sound modern today?

Prepare for class

- Read the assigned sections
- Write a half-page note per paper
- Bring one answer to each question
- Discuss in groups of three

DS 501 · WEEK 1

This Week's Work

Practice 1 notebook

Finish W01_Lab_First_Look_at_Data.ipynb, check that it runs from top to bottom, and submit the .ipynb file.

Due: [date]

Quiz 1

15 multiple-choice questions on lessons 1-1 to 1-3. Review the objectives and Key Terms slides first.

Due: [date]

Project step 1

Choose a domain. Write one paragraph on a decision in it and list 2–3 datasets that could inform it.

Due: [date]

Next week: Data Science Methodology: how a project moves from a business problem to a deployed model, step by step, and how CRISP-DM organises the work.